

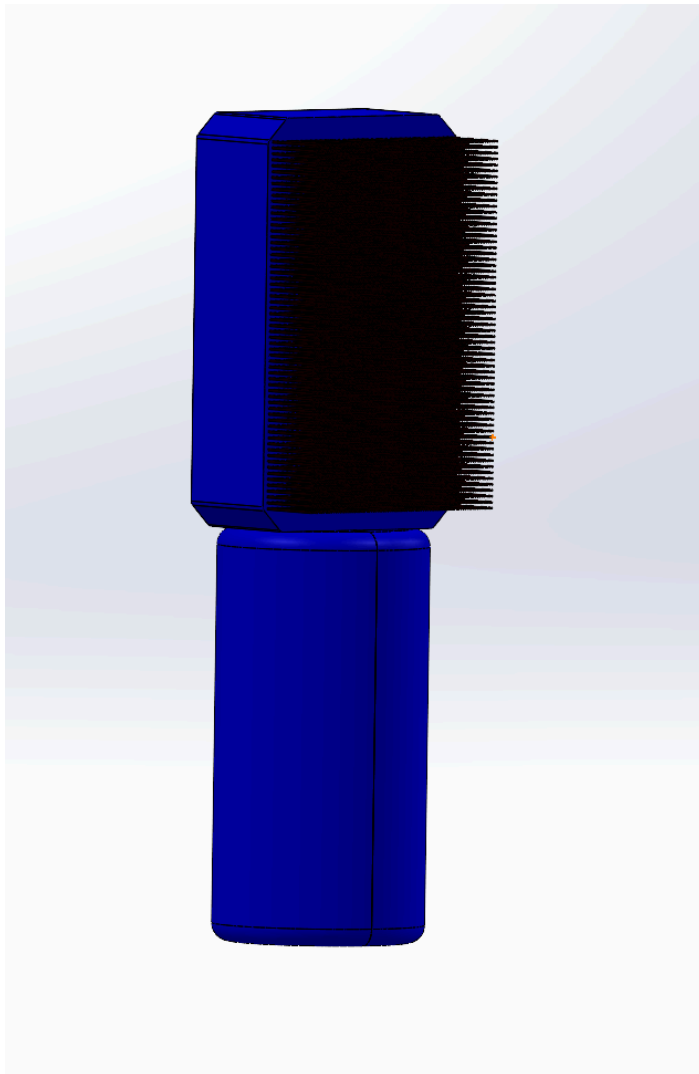
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Title: *Multi-Purpose Pet Brush – Recycled Plastic Design*

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Title and Description

The **Multi-Purpose Pet Brush** is designed to make animal grooming easier and more organized. It features a hinged handle with internal storage compartments, a removable brush head for different fur types, and space for carrying grooming accessories such as nail clippers or waste bags. The device was modeled using SolidWorks, with six primary parts created and assembled using standard and advanced mates.

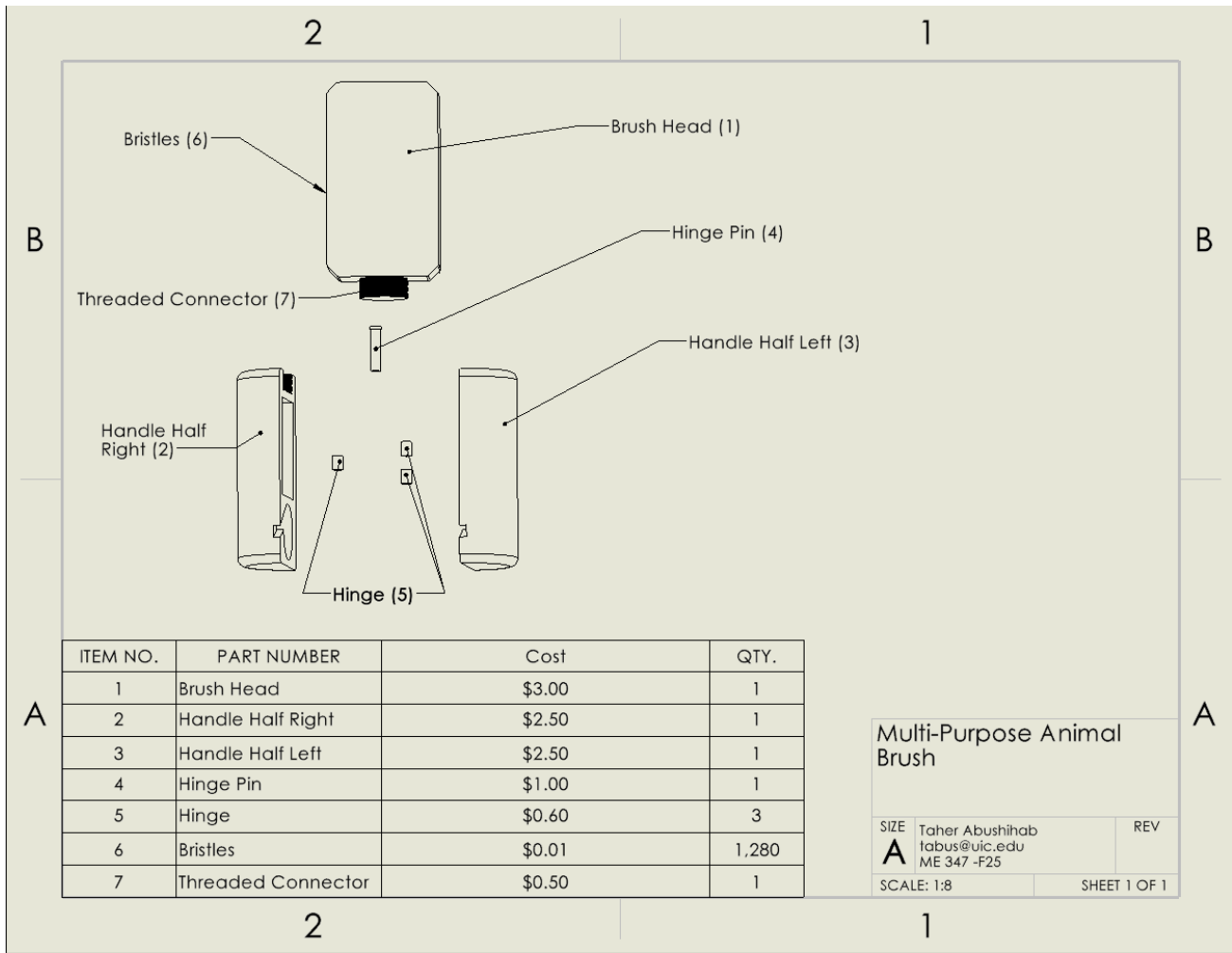
Parts and Cost

The Multi-Purpose Animal Brush consists of seven key components: the brush head, two handle halves, a hinge pin, three hinges, a large array of bristles, and a threaded connector that secures the brush head to the handle. To estimate manufacturing cost, each part's material, size, and expected production method were considered. Prices were referenced using common market rates for small-scale plastic injection-molded and metal parts found on supplier catalogs such as McMaster-Carr and Alibaba for bulk orders.

The brush head and handle halves were assumed to be made from recycled PET plastic, which typically costs around \$2.50–\$3.00 per unit at low-volume runs. The hinge pin and hinges are small precision parts; their cost was approximated from stainless-steel hinge sets at roughly \$0.60–\$1.00 each. The bristles, modeled as patterned features, represent mass-produced flexible nylon bristles that cost approximately \$0.01 per unit when purchased in high quantities (over 1,000 pieces). The threaded connector, a small cylindrical insert allowing the brush head to be swapped, was estimated at \$0.50 based on typical molded insert pricing.

Overall, the total estimated material cost per assembled unit is approximately \$10–\$12, excluding labor and packaging. This price range aligns with entry-level grooming brushes available commercially, supporting the product's feasibility for low-cost, sustainable production.

Exploded View Drawing



Design Improvement

The final version of the Multi-Purpose Animal Brush focuses on refining manufacturability and assembly rather than making major design overhauls. The most significant improvement was the addition of threaded connections between the brush head and handle. This change allows users to easily detach and replace brush heads for cleaning or switching between fur-type attachments. The threaded connector also provides a tighter and more reliable fit compared to earlier press-fit concepts, reducing wear over time.

Minor adjustments were made to improve alignment and tolerances between the handle halves, ensuring the hinge fits correctly and the halves close without gaps. The internal geometry of the handle was slightly simplified to make it easier to mold and assemble while maintaining enough internal space for small pet accessories such as waste bags. Additionally, bristle density and arrangement were refined to improve grooming performance and achieve a realistic pattern that could be manufactured using standard bristle molds.

Although the overall form of the brush remained consistent with the original concept, these small refinements made the design cleaner, more functional, and better suited for real-world manufacturing and use.

Performance Evaluation

The Multi-Purpose Pet Brush performs effectively as both a grooming and storage device when evaluated based on the project objectives. The hinge assembly functions smoothly, allowing the lid to open and close without interference, while the internal compartment provides adequate space for small grooming accessories such as nail clippers or waste bags. The brush heads attach securely to the handle, demonstrating proper alignment and ease of removal, which highlights successful application of mates and part relations in SolidWorks.

From a usability standpoint, the overall shape of the handle provides a comfortable grip, and the placement of the hinge and storage compartment ensures that the design remains balanced and compact. The bristle pattern on the brush heads, created using a linear feature pattern, provides full surface coverage for efficient grooming of different pet fur types. In terms of durability, the combination of recycled PET for the main body and stainless steel for the hinge pin ensures long-term functionality under repeated use.

Finally, the design also meets sustainability and manufacturability goals. Each part can be injection-molded or 3D printed with minimal post-processing, making the product efficient to produce. The use of recycled materials further reduces waste and cost. Overall, the final assembly meets all intended performance criteria—functionality, comfort, strength, and environmental responsibility—while successfully demonstrating advanced SolidWorks design and assembly techniques.

Conclusion

The Multi-Purpose Pet Brush project allowed me to apply SolidWorks to a complete, realistic design problem. Starting from initial sketches, I developed a fully assembled product that combines multiple functions—storage, interchangeable brush heads, and an ergonomic recycled-plastic body.

Through this process, I strengthened my understanding of part modeling, mates, and motion constraints while learning to think critically about manufacturability and sustainability. The final design reflects a balance of practicality, durability, and environmental responsibility, demonstrating how thoughtful engineering can make everyday products both functional and eco-friendly.